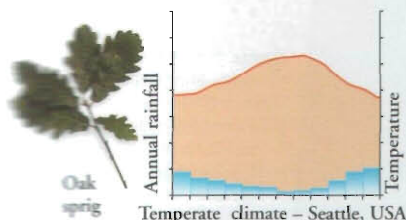


CLIMATE



WEATHER CAN CHANGE from moment to moment. Over a long period of time, a region's characteristic weather – however changeable – is called its climate. Climates are generally warm near the Equator, the imaginary line around the middle of the Earth, and cool towards the poles. Other influences on the three broad climate types – warm tropical, cold polar, and mild temperate – include the distance from sea and the position within a continent. The climate determines a region's animal and plant life.



Temperate climate

In mid-latitude (imaginary lines parallel to the Equator) areas such as the USA, summers are warm, and winters cool, with regular rain. A Mediterranean climate with dry summers and warm, damp winters is a type of temperate climate.



Mountain climate

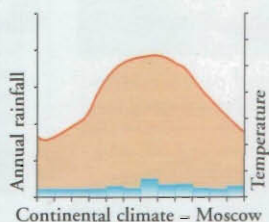
High altitude causes the air to cool, thus creating a cold climate. Exposed mountain tops also make mountain climates very wet and windy. Above a certain height called the snow-line, there is always snow.



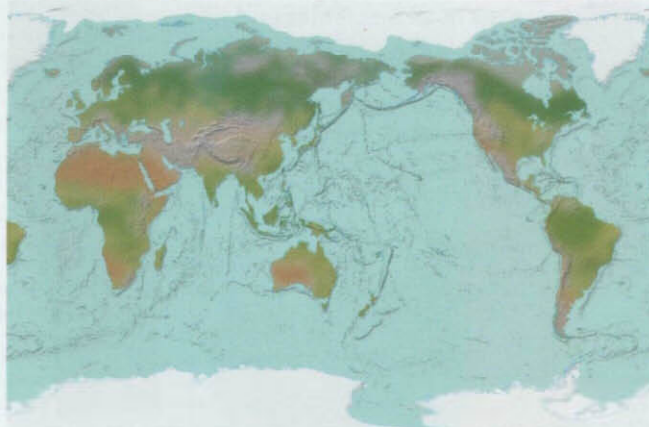
Each tree ring shows one year's growth: a wide ring means the weather was warm and the tree grew well.

Oceanic and continental climate

Coastal regions have wet, changeable weather. The summers are cooler and the winters are warmer, because the ocean heats up and cools down more slowly than the land. Places in the continental interior, such as Moscow, have cold winters.



Oceanic and continental climate zones



Polar
Mountain
Tundra
Temperate
Mediterranean
Dry grassland
Desert
Subtropical
Tropical



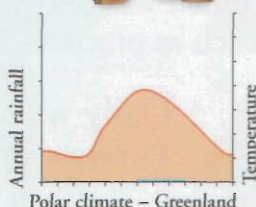
Climate zones

Close to the Equator, the Sun's warmth is strong – it climbs high in the sky at midday: closer to the poles, the Sun's warmth is weaker – it climbs less high.

Climatic zones, which effect vegetation, can be further classified by physical features.



Polar bear



Polar climate

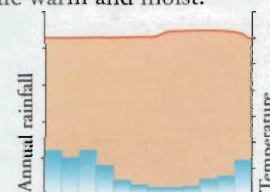
Towards the ice-capped poles, the Sun is always low in the sky, and in winter barely rises at all: summers are brief. Winter temperatures in the tundra regions around the North Pole are below -60°C (-76°F).

Tropical climate

Weather in a tropical climate, such as Brazil, is always warm, often with heavy rainfall. Some tropical climates, such as deserts, are hot and dry; others, such as rain-forests, are warm and moist.



Parrots



Tropical climate – Brazil

Monsoons

These are warm, tropical climates with wet and dry seasons. In India, it is dry from October to May as the winds blow out to sea, and very wet from June to September as the monsoon winds blow inland.



Desert climate

Over a fifth of the world's land surface is desert, where there is typically an annual rainfall of less than 100 mm (4 in). In the tropics, desert temperatures frequently climb above 50°C (122°F).



Global warming

Pollution may be warming the world up. Certain gases trap the Sun's heat in the Earth's atmosphere. Rising levels of these "greenhouse gases", such as carbon dioxide, which come from burning oil or forest land, may trap so much heat that the Earth could warm up by 4°C (7.2°F) over the next 50 years.



FIND OUT
MORE

DESERTS

MOUNTAINS
AND VALLEYS

OCEANS
AND SEAS

POLLUTION

RAIN

WEATHER

WEATHER
FORECASTING

WINDS